

Block Ciphers - DES & AES.

Date:

- Types of Symmetric Ciphers.
DES & AES are block Ciphers.
 - Block cipher (Process fixed-sized blocks of plain text) (64 or 128 bits)
 - Stream Cipher. (Encrypt data bit by bit (or byte by byte).
→ faster + more sensitive to key reuse.
- Block Cipher Principles
 - Most Symmetric block Ciphers are based on Feistel Cipher Structure.
 - Should be reversible
 - If each block n bits long, 2^n possible combinations of bits.
- Block Cipher are Strong & Secure
 - Confusion → • makes relationship b/w encryption key & ciphertext complex & non linear.
 - If attacker know part of ciphertext, it prevents.
 - In DES → S-boxes add this non-linearity.
 - Diffusion → • Spread plaintext statistics across ciphertext.
 - Avalanche effect → ensures changing 1 bit of plaintext changes many bits of ciphertext.
 - In DES → S-box permutations + multiple rounds spread the influence.

- provides simple way to build secure encryption & decryption algorithms.

→ **Fiestel Cipher** → not a complete cipher itself but a design model used to build many block ciphers e.g., DES.

- The differences During decryption, round key are applied in reverse order.

Features → • Works on block of data

- In Plaintext block is split into two halves: Left(L) & Right(R)
- In each round, one half is modified using a function of the other half and a round key
- Multiple rounds make cipher stronger.
- Same process used for both encryption & decryption.

→ Steps:-

i) Take plaintext & convert it to ASCII & then to Binary.

$$A = 65 = 01000001$$

$$B = 66 = 01000010$$

$$• 0100000101000010$$

ii) Divide into 2 halves.

$$\text{Left (L)} = 01000001$$

$$\text{Right (R)} = 01000010$$

iii) Create 2 random Binary keys.

$$K_1 = 10101010$$

$$K_2 = 11001100$$

XOR

0	0	0
0	1	1
1	0	1
1	1	0

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Encryption Process (2 Fiestel rounds)

Round 1

$$\begin{aligned} f_1 &= R_1 \text{ XOR } K_1 \\ R_2 &= L_1 \text{ XOR } f_1 \\ L_2 &= R_1 \end{aligned}$$

Round 2

$$\begin{aligned} f_2 &= R_2 \text{ XOR } K_2 \\ R_3 &= L_2 \text{ XOR } f_2 \\ L_3 &= R_2 \end{aligned}$$

Final Ciphertext.

Concatenate L_3 & R_3 for Cipher text.

Decryption Process (same but round keys applied in reverse order)

⇒ DES - Function of Fiestel structure.

i) • We have 64 bit blocks.

H 01001000

e 01100101

l 01101100

l 01101100

o 01101111

00100000

W 01010111

O 01101111

r 01110010

l 01101100

d 01100100

l 00100001

Input: 64 bits
Output: 64 bits

Main Key: 64 bits
Sub Key: 16 bits

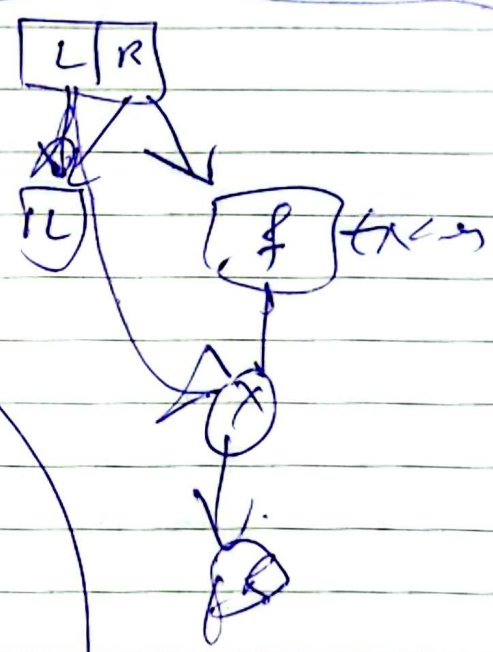
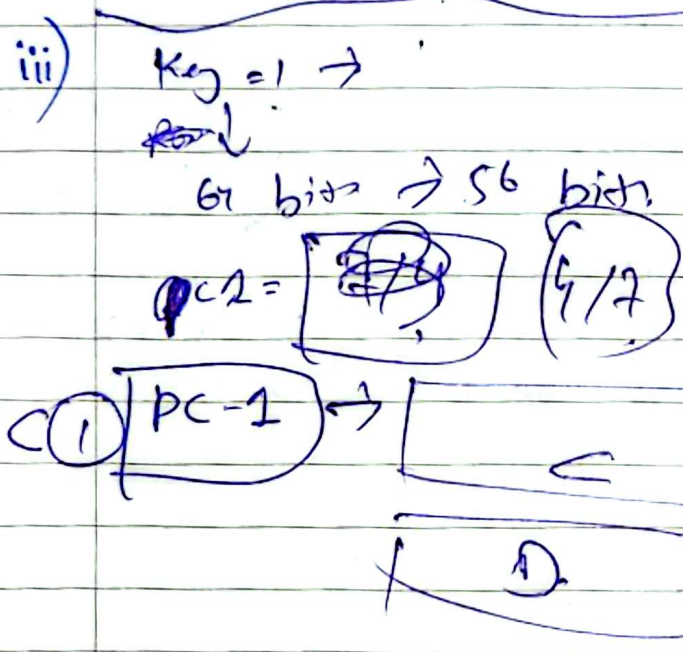
Round Key: 48 bits
No. of rounds: 16 rounds

It is not matrix. Shown in Date:
↑ rows & cols for readability

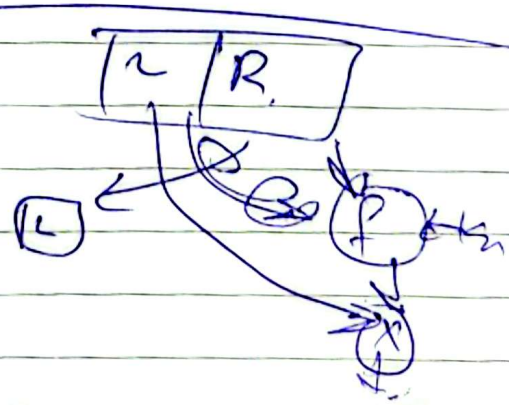
- ii) Split entire input into blocks 64 bit each.
• Break into 64 bit blocks.

b₁ =
0 1 0 0 1 0 0 0
0 1 1 0 0 1 0 1
0 1 1 0 1 1 0 0
0 1 1 0 1 1 0 0
0 1 1 0 1 1 1 1
0 0 1 0 0 0 0 0
0 1 0 1 0 1 1 1
0 1 1 0 1 1 1 1

b₂ =
0 1 1 1 0 0 1 0
0 1 1 0 1 1 0 0
0 1 1 0 0 1 0 0
0 0 1 0 0 0 0 1
Padding
7/4, Padding
& Padding
CED Padding.



(2) Circular Shift. 1 2 3 4
2 3 4 1
R 1, 2, 9, 16 → Left Shift 1



(3) S-box. (4) PC-2

1) Initial Permutation (IP)

1	2	3	4	5	6	7	8	58	50	42	34	26	18	10	2
9	10	11	12	13	14	15	16	60	52	44	36	28	20	12	4
17	18	19	20	21	22	23	24	62	54	46	38	30	22	14	6
25	26	27	28	29	30	31	32	64	56	48	40	32	24	16	8
33	34	35	36	37	38	39	40	57	49	41	33	25	17	9	1
41	42	43	44	45	46	47	48	59	51	43	35	27	19	11	3
49	50	51	52	53	54	55	56	61	53	45	37	29	21	13	5
57	58	59	60	61	62	63	64	63	55	47	39	31	23	15	7



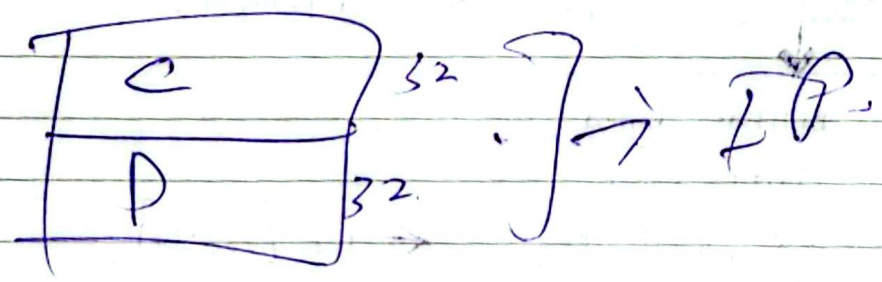
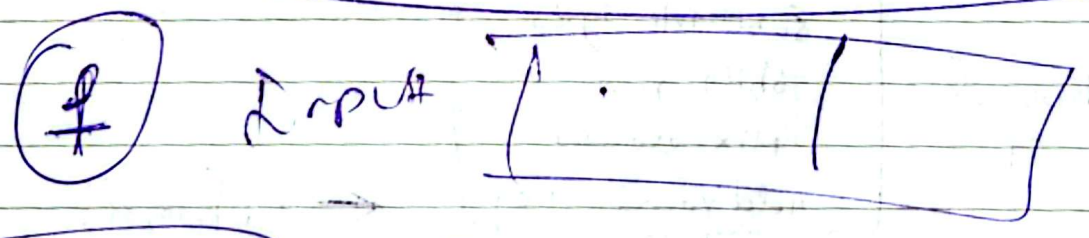
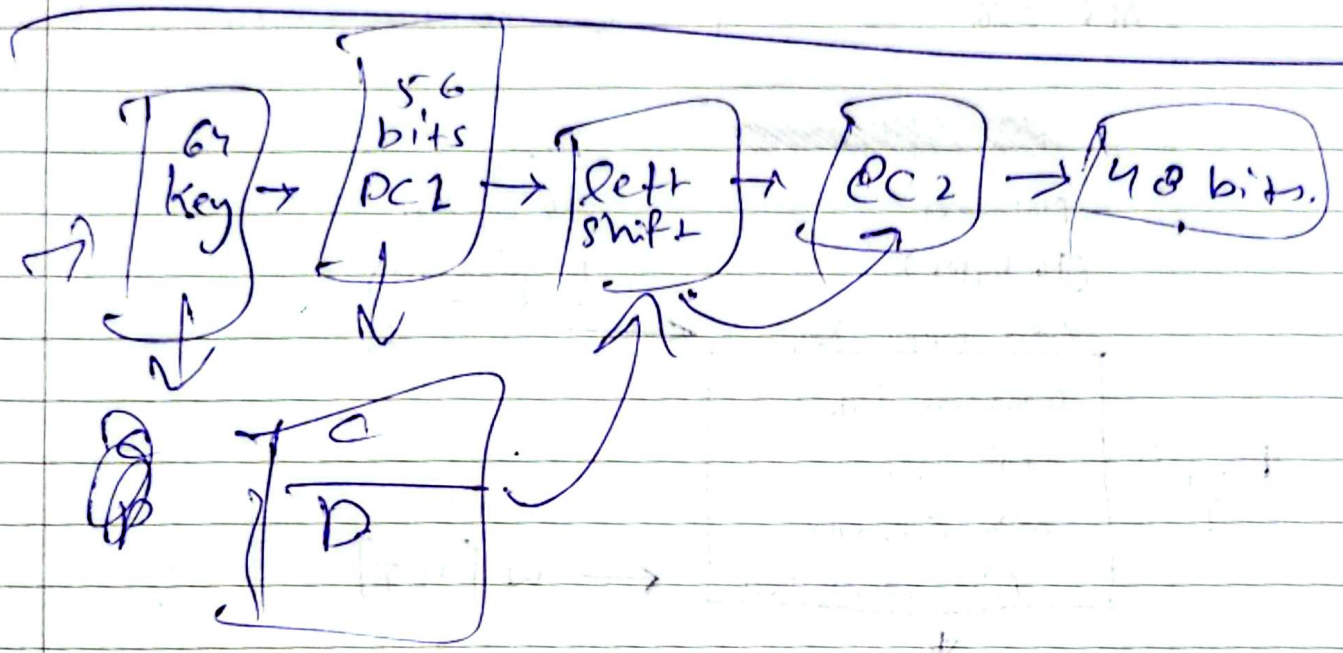
2) Inverse Initial Permutation.

40	8	48	16	56	24	64	32
39	7	47	15	55	23	63	31
38	6	46	14	54	22	62	30
37	5	45	13	53	21	61	29
36	4	44	12	52	20	60	28
35	3	43	11	51	19	59	27
34	2	42	10	50	18	58	26
33	1	41	9	49	17	57	25

2nd row $\rightarrow$ 1st row
 4th $\rightarrow$ 2nd
 8 $\rightarrow$ 3rd
 8 $\rightarrow$ 4th
 1 $\rightarrow$ 5th
 3 $\rightarrow$ 6th

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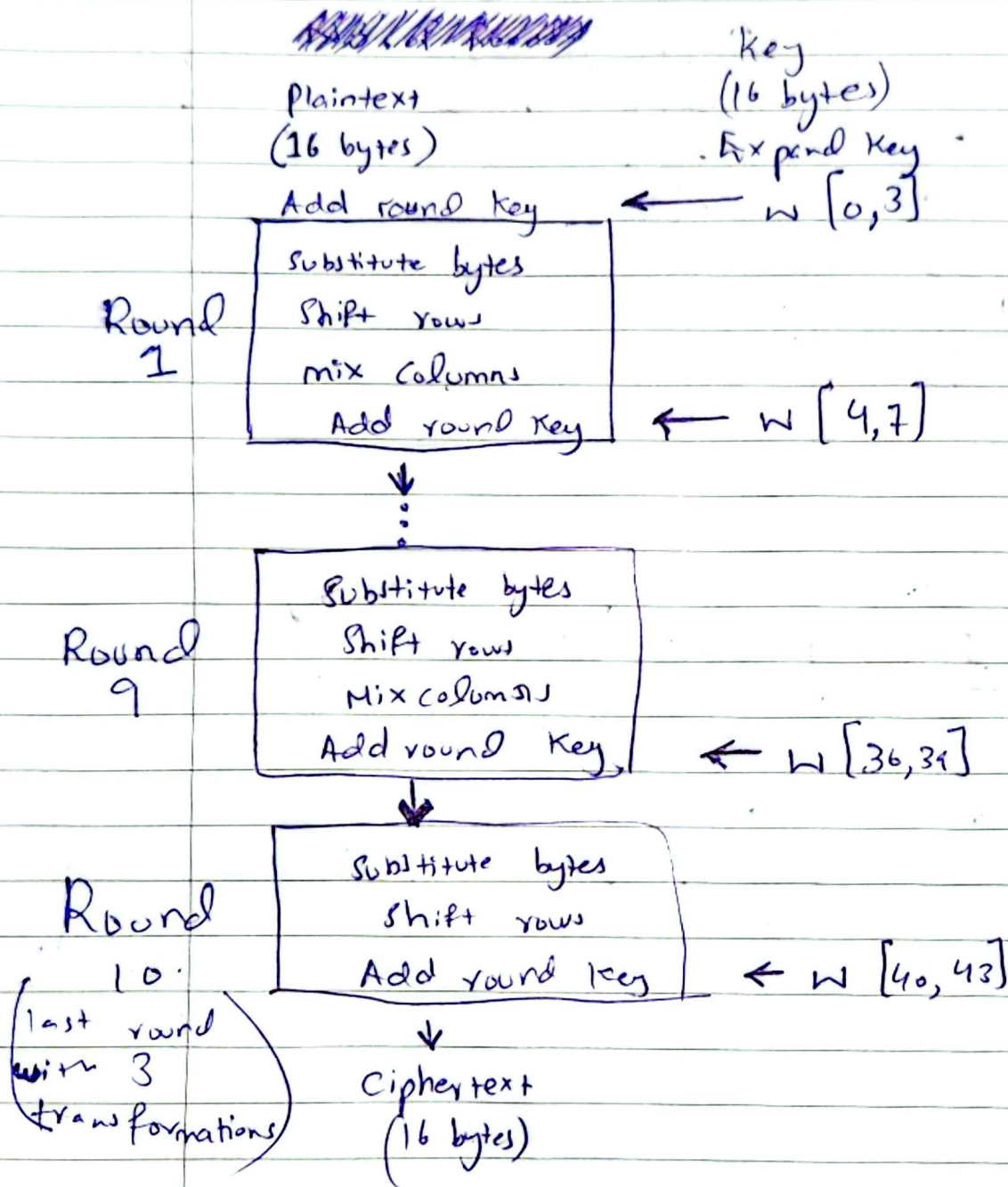
3) Single Round of DES Algorithm.
Round 1 (Takes 64 bits input + 48 bits key) (Then 64 bits as output).



Symmetric Block cipher

1) AES (Advance Encryption Standard) Date: _____

	key length	Block size	Round key size	no of rounds	Plaintext size
AES-128	4	4	128	10	128
AES-192	6	4	128	12	128
AES-256	8	4	128	14	128



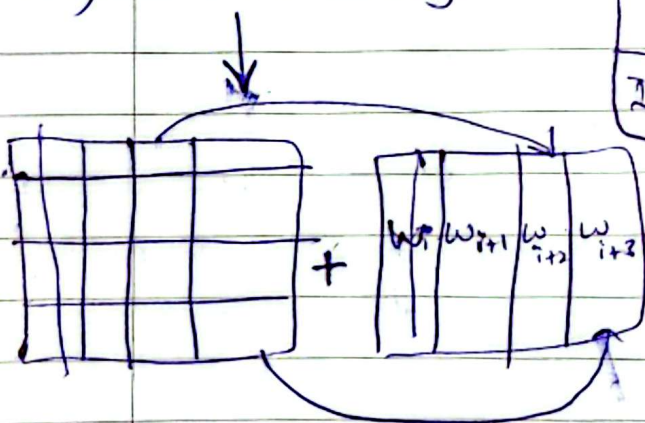
- 1) Substitute Bytes (S-Box - 16×16 matrix)
- 2) Shift Rows.

S_{00}	S_{01}	S_{02}	S_{03}		S_{00}	S_{01}	S_{02}	S_{03}
S_{10}	S_{11}	S_{12}	S_{13}	$\rightarrow$	S_{11}	S_{12}	S_{13}	S_{10}
S_{20}	S_{21}	S_{22}	S_{23}		S_{22}	S_{23}	S_{20}	S_{21}
S_{30}	S_{31}	S_{32}	S_{33}		S_{33}	S_{30}	S_{32}	S_{31}

- 3) Mix Columns.

S_{00}	S_{01}	S_{02}	S_{03}	$\rightarrow$ Multiply with a matrix $\rightarrow$	S'	S'	S'	S'
S_{10}	S_{11}	S_{12}	S_{13}		S'	S'	S'	S'
S_{20}	S_{21}	S_{22}	S_{23}		S'	S'	S'	S'
S_{30}	S_{31}	S_{32}	S_{33}		S'	S'	S'	S'

- 4) Round Key.



Rounds	words.
Initial Transformation	w_0, w_1, w_2, w_3
Round 1	w_4, w_5, w_6, w_7
Round 2	w_8, w_9, w_{10}, w_{11}
...	
Round 10	$w_{40}, w_{41}, w_{42}, w_{43}$

AES Key Expansion

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K_0	K_4	K_8	K_{12}
K_1	K_5	K_9	K_{13}
K_2	K_6	K_{10}	K_{14}
K_3	K_7	K_{11}	K_{15}

